



Intel® 64 and IA-32 Architectures Software Developer's Manual

Combined Volumes:
1, 2A, 2B, 2C, 2D, 3A, 3B, 3C, 3D, and 4

NOTE: This document contains all four volumes of the Intel 64 and IA-32 Architectures Software Developer's Manual: *Basic Architecture*, Order Number 253665; *Instruction Set Reference A-Z*, Order Number 325383; *System Programming Guide*, Order Number 325384; *Model-Specific Registers*, Order Number 335592. Refer to all four volumes when evaluating your design needs.

Order Number: 325462-079US
March 2023

Notices & Disclaimers

Intel technologies may require enabled hardware, software or service activation.
No product or component can be absolutely secure.

Your costs and results may vary.

You may not use or facilitate the use of this document in connection with any infringement or other legal analysis concerning Intel products described herein. You agree to grant Intel a non-exclusive, royalty-free license to any patent claim thereafter drafted which includes subject matter disclosed herein.

All product plans and roadmaps are subject to change without notice.

The products described may contain design defects or errors known as errata which may cause the product to deviate from published specifications. Current characterized errata are available on request.

Intel disclaims all express and implied warranties, including without limitation, the implied warranties of merchantability, fitness for a particular purpose, and non-infringement, as well as any warranty arising from course of performance, course of dealing, or usage in trade.

Code names are used by Intel to identify products, technologies, or services that are in development and not publicly available. These are not “commercial” names and not intended to function as trademarks.

No license (express or implied, by estoppel or otherwise) to any intellectual property rights is granted by this document, with the sole exception that a) you may publish an unmodified copy and b) code included in this document is licensed subject to the Zero-Clause BSD open source license (0BSD), <https://opensource.org/licenses/0BSD>. You may create software implementations based on this document and in compliance with the foregoing that are intended to execute on the Intel product(s) referenced in this document. No rights are granted to create modifications or derivatives of this document.

© Intel Corporation. Intel, the Intel logo, and other Intel marks are trademarks of Intel Corporation or its subsidiaries. Other names and brands may be claimed as the property of others.



Intel® 64 and IA-32 Architectures Software Developer's Manual

Volume 1: Basic Architecture

NOTE: The Intel 64 and IA-32 Architectures Software Developer's Manual consists of four volumes: *Basic Architecture*, Order Number 253665; *Instruction Set Reference A-Z*, Order Number 325383; *System Programming Guide*, Order Number 325384; *Model-Specific Registers*, Order Number 335592. Refer to all four volumes when evaluating your design needs.

Notices & Disclaimers

Intel technologies may require enabled hardware, software or service activation.

No product or component can be absolutely secure.

Your costs and results may vary.

You may not use or facilitate the use of this document in connection with any infringement or other legal analysis concerning Intel products described herein. You agree to grant Intel a non-exclusive, royalty-free license to any patent claim thereafter drafted which includes subject matter disclosed herein.

All product plans and roadmaps are subject to change without notice.

The products described may contain design defects or errors known as errata which may cause the product to deviate from published specifications. Current characterized errata are available on request.

Intel disclaims all express and implied warranties, including without limitation, the implied warranties of merchantability, fitness for a particular purpose, and non-infringement, as well as any warranty arising from course of performance, course of dealing, or usage in trade.

Code names are used by Intel to identify products, technologies, or services that are in development and not publicly available. These are not “commercial” names and not intended to function as trademarks.

No license (express or implied, by estoppel or otherwise) to any intellectual property rights is granted by this document, with the sole exception that a) you may publish an unmodified copy and b) code included in this document is licensed subject to the Zero-Clause BSD open source license (0BSD), <https://opensource.org/licenses/0BSD>. You may create software implementations based on this document and in compliance with the foregoing that are intended to execute on the Intel product(s) referenced in this document. No rights are granted to create modifications or derivatives of this document.

© Intel Corporation. Intel, the Intel logo, and other Intel marks are trademarks of Intel Corporation or its subsidiaries. Other names and brands may be claimed as the property of others.

CONTENTS

PAGE

CHAPTER 1

ABOUT THIS MANUAL

1.1 INTEL [®] 64 AND IA-32 PROCESSORS COVERED IN THIS MANUAL	1-1
1.2 OVERVIEW OF VOLUME 1: BASIC ARCHITECTURE	1-4
1.3 NOTATIONAL CONVENTIONS	1-6
1.3.1 Bit and Byte Order.	1-6
1.3.2 Reserved Bits and Software Compatibility	1-6
1.3.2.1 Instruction Operands.	1-7
1.3.3 Hexadecimal and Binary Numbers	1-7
1.3.4 Segmented Addressing	1-7
1.3.5 A New Syntax for CPUID, CR, and MSR Values	1-7

.....1-8 1.3.6 Exceptions	1-8
.....1-8 1.4 RELATED LITERATURE	1-9
CHAPTER 2	
INTEL® 64 AND IA-32 ARCHITECTURES	
2.1 BRIEF HISTORY OF INTEL® 64 AND IA-32 ARCHITECTURE	
..... 2-1 2.1.1 16-bit Processors and Segmentation (1978)	
..... 2-1 2.1.2 The Intel® 286 Processor (1982)	
..... 2-1 2.1.3 The Intel386™ Processor (1985)	
..... 2-1 2.1.4 The Intel486™ Processor (1989)	
..... 2-1 2.1.5 The Intel® Pentium® Processor (1993)	
..... 2-2 2.1.6 The P6 Family of Processors (1995-1999)	
..... 2-2 2.1.7 The Intel® Pentium® 4 Processor Family (2000-2006)	
..... 2-3 2.1.8 The Intel® Xeon® Processor (2001- 2007)	
..... 2-3 2.1.9 The Intel® Pentium® M Processor (2003-2006)	
..... 2-3 2.1.10 The Intel® Pentium® Processor Extreme Edition (2005)	
..... 2-4 2.1.11 The Intel® Core™ Duo and Intel® Core™ Solo Processors (2006-2007)	
..... 2-4 2.1.12 The Intel® Xeon® Processor 5100, 5300 Series and Intel® Core™ 2 Processor Family (2006)	
..... 2-4 2.1.13 The Intel® Xeon® Processor 5200, 5400, 7400 Series and Intel® Core™ 2 Processor Family (2007)	
..... 2-4 2.1.14 The Intel Atom® Processor Family (2008)	
..... 2-5 2.1.15 The Intel Atom® Processor Family Based on Silvermont Microarchitecture (2013)	
..... 2-5 2.1.16 The Intel® Core™ i7 Processor Family (2008)	
..... 2-5 2.1.17 The Intel® Xeon® Processor 7500 Series (2010)	
..... 2-5 2.1.18 2010 Intel® Core™ Processor Family (2010)	
..... 2-6 2.1.19 The Intel® Xeon® Processor 5600 Series (2010) ..	
..... 2-6 2.1.20 The Second Generation Intel® Core™ Processor Family (2011)	
..... 2-6 2.1.21 The Third Generation Intel® Core™ Processor Family (2012)	
..... 2-6 2.1.22 The Fourth Generation Intel® Core™ Processor Family (2013)	
..... 2-7 2.2 MORE ON SPECIFIC ADVANCES	
..... 2-7 2.2.1 P6 Family Microarchitecture	
..... 2-7 2.2.2 Intel NetBurst® Microarchitecture	
..... 2-8 2.2.2.1 The Front End Pipeline	
..... 2-9 2.2.2.2 Out-Of-Order Execution Core	
..... 2-10 2.2.2.3 Retirement Unit	
..... 2-10 2.2.3 Intel® Core™ Microarchitecture	
..... 2-10 2.2.3.1 The Front End	
..... 2-11 2.2.3.2 Execution Core	
..... 2-12 2.2.4 Intel Atom® Microarchitecture	
..... 2-12 2.2.5 Nehalem Microarchitecture	
..... 2-13 2.2.6 Sandy Bridge Microarchitecture	
..... 2-13 2.2.7 SIMD Instructions	
..... 2-14 2.2.8 Intel® Hyper-Threading Technology	
..... 2-16 2.2.8.1 Some Implementation Notes	
..... 2-17 2.2.9 Multi-Core Technology	
..... 2-18	

CONTENTS

PAGE

2.2.10 Intel® 64 Architecture	
..... 2-20 2.2.11 Intel® Virtualization Technology (Intel® VT)	
..... 2-20 2.3 INTEL® 64 AND IA-32 PROCESSOR GENERATIONS	
..... 2-20 2.4 PLANNED REMOVAL OF INTEL® INSTRUCTION SET ARCHITECTURE AND FEATURES FROM UPCOMING PRODUCTS.	
..... 2-28 2.5 INTEL® INSTRUCTION SET ARCHITECTURE AND FEATURES REMOVED	2-28

CHAPTER 3

BASIC EXECUTION ENVIRONMENT

3.1 MODES OF OPERATION	3-1
3.1.1 Intel® 64 Architecture	3-1
3.2 OVERVIEW OF THE BASIC EXECUTION ENVIRONMENT	3-2
3.2.1 64-Bit Mode Execution Environment	3-5
3.3 MEMORY ORGANIZATION	3-6
3.3.1 IA-32 Memory Models	3-7
3.3.2 Paging and Virtual Memory	3-8
3.3.3 Memory Organization in 64-Bit Mode	3-8
3.3.4 Modes of Operation vs. Memory Model	3-9
3.3.5 32-Bit and 16-Bit Address and Operand Sizes	3-9
3.3.6 Extended Physical Addressing in Protected Mode	3-9
3.3.7 Address Calculations in 64-Bit Mode	3-10
3.3.7.1 Canonical Addressing	3-10
3.4 BASIC PROGRAM EXECUTION REGISTERS	3-10
3.4.1 General-Purpose Registers	3-11
3.4.1.1 General-Purpose Registers in 64-Bit Mode	3-12
3.4.2 Segment Registers	3-13
3.4.2.1 Segment Registers in 64-Bit Mode	3-15
3.4.3 EFLAGS Register	3-15
3.4.3.1 Status Flags	3-16
3.4.3.2 DF Flag	3-17
3.4.3.3 System Flags and IOPL Field	3-17
3.4.3.4 RFLAGS Register in 64-Bit Mode	3-18
3.5 INSTRUCTION POINTER	3-18
3.5.1 Instruction Pointer in 64-Bit Mode	3-18
3.6 OPERAND-SIZE AND ADDRESS-SIZE ATTRIBUTES	3-18
3.6.1 Operand Size and Address Size in 64-Bit Mode	3-19
3.7 OPERAND ADDRESSING	3-19
3.7.1 Immediate Operands	3-20
3.7.2 Register Operands	3-20
3.7.2.1 Register Operands in 64-Bit Mode	3-21
3.7.3 Memory Operands	3-21
3.7.3.1 Memory Operands in 64-Bit Mode	3-21
3.7.4 Specifying a Segment Selector	3-21
3.7.4.1 Segmentation in 64-Bit Mode	3-22
3.7.5 Specifying an Offset	3-22
3.7.5.1 Specifying an Offset in 64-Bit Mode	3-24
3.7.6 Assembler and Compiler Addressing Modes	3-24
3.7.7 I/O Port Addressing	3-24

CHAPTER 4

DATA TYPES

4.1 FUNDAMENTAL DATA TYPES	4-1
4.1.1 Alignment of Words, Doublewords, Quadwords, and Double Quadwords	4-2
4.2 NUMERIC DATA TYPES	4-2
4.2.1 Integers	4-3
4.2.1.1 Unsigned Integers	4-3
4.2.1.2 Signed Integers	4-4
4.2.2 Floating-Point Data Types	4-4
4.3 POINTER DATA TYPES	4-6
4.3.1 Pointer Data Types in 64-Bit Mode	4-7

4.4 BIT FIELD DATA TYPE	4-7
4.5 STRING DATA TYPES	4-8
4.6 PACKED SIMD DATA TYPES	4-8
4.6.1 64-Bit SIMD Packed Data Types	4-8
4.6.2 128-Bit Packed SIMD Data Types	4-8
4.7 BCD AND PACKED BCD INTEGERS	4-8
4.8 REAL NUMBERS AND FLOATING-POINT FORMATS	4-9
4.8.1 Real Number System	4-11
4.8.2 Floating-Point Format	4-11
4.8.2.1 Normalized Numbers	4-11
4.8.2.2 Biased Exponent	4-13
4.8.3 Real Number and Non-number Encodings	4-13
4.8.3.1 Signed Zeros	4-13
4.8.3.2 Normalized and Denormalized Finite Numbers	4-14
4.8.3.3 Signed Infinities	4-14
4.8.3.4 NaNs	4-15
4.8.3.5 Operating on SNaNs and QNaNs	4-15
4.8.3.6 Using SNaNs and QNaNs in Applications	4-16
4.8.3.7 QNaN Floating-Point Indefinite	4-16
4.8.3.8 Half Precision Floating-Point Operation	4-17
4.8.4 Rounding	4-17
4.8.4.1 Rounding Control (RC) Fields	4-17
4.8.4.2 Truncation with Intel® SSE, SSE2, and AVX Conversion Instructions	4-18
4.9 OVERVIEW OF FLOATING-POINT EXCEPTIONS	4-18
4.9.1 Floating-Point Exception Conditions	4-19
4.9.1.1 Invalid Operation Exception (#I)	4-20
4.9.1.2 Denormal Operand Exception (#D)	4-20
4.9.1.3 Divide-By-Zero Exception (#Z)	4-20
4.9.1.4 Numeric Overflow Exception (#O)	4-21
4.9.1.5 Numeric Underflow Exception (#U)	4-21
4.9.1.6 Inexact-Result (Precision) Exception (#P)	4-22
4.9.2 Floating-Point Exception Priority	4-22
4.9.3 Typical Actions of a Floating-Point Exception Handler	4-23

CHAPTER 5

INSTRUCTION SET SUMMARY

5.1 GENERAL-PURPOSE INSTRUCTIONS	5-5
5.1.1 Data Transfer Instructions	5-5
5.1.2 Binary Arithmetic Instructions	5-6
5.1.3 Decimal Arithmetic Instructions	5-6
5.1.4 Logical Instructions	5-7
5.1.5 Shift and Rotate Instructions	5-7
5.1.6 Bit and Byte Instructions	5-7
5.1.7 Control Transfer Instructions	5-7
5.1.8 String Instructions	5-8
5.1.9 I/O Instructions	5-9
5.1.10 Enter and Leave Instructions	5-9
5.1.11 Flag Control (EFLAG) Instructions	5-10
5.1.12 Segment Register Instructions	5-10
5.1.13 Miscellaneous Instructions	5-10

5-10 5.1.14 User Mode Extended State Save/Restore Instructions	
5-11 5.1.15 Random Number Generator Instructions	
5-11 5.1.16 BMI1 and BMI2 Instructions	
5-11 5.1.16.1 Detection of VEX-Encoded GPR Instructions, LZCNT, TZCNT, and PREFETCHW	
5-11 5.2 X87 FPU INSTRUCTIONS	
5-12 5.2.1 x87 FPU Data Transfer Instructions	
5-12 5.2.2 x87 FPU Basic Arithmetic Instructions	
5-12 5.2.3 x87 FPU Comparison Instructions	
5-13 5.2.4 x87 FPU Transcendental Instructions.	
5-13	

CONTENTS

PAGE

5.2.5 x87 FPU Load Constants Instructions	
5-14 5.2.6 x87 FPU Control Instructions	
5-14 5.3 X87 FPU AND SIMD STATE MANAGEMENT INSTRUCTIONS	
5-14 5.4 MMX INSTRUCTIONS	
5-15 5.4.1 MMX Data Transfer Instructions	
5-15 5.4.2 MMX Conversion Instructions.	
5-15 5.4.3 MMX Packed Arithmetic Instructions	
5-15 5.4.4 MMX Comparison Instructions	
5-16 5.4.5 MMX Logical Instructions.	
5-16 5.4.6 MMX Shift and Rotate Instructions	
5-16 5.4.7 MMX State Management Instructions	
5-16 5.5 INTEL® SSE INSTRUCTIONS	
5-17 5.5.1 Intel® SSE SIMD Single Precision Floating-Point Instructions	
5-17 5.5.1.1 Intel® SSE Data Transfer Instructions	
5-17 5.5.1.2 Intel® SSE Packed Arithmetic Instructions	
5-17 5.5.1.3 Intel® SSE Comparison Instructions.	
5-18 5.5.1.4 Intel® SSE Logical Instructions	
5-18 5.5.1.5 Intel® SSE Shuffle and Unpack Instructions	
5-18 5.5.1.6 Intel® SSE Conversion Instructions	
5-18 5.5.2 Intel® SSE MXCSR State Management Instructions	
5-19 5.5.3 Intel® SSE 64-Bit SIMD Integer Instructions	
5-19 5.5.4 Intel® SSE Cacheability Control, Prefetch, and Instruction Ordering Instructions.	
5-19 5.6 INTEL® SSE2 INSTRUCTIONS.	
5-19 5.6.1 Intel® SSE2 Packed and Scalar Double Precision Floating-Point Instructions	
5-20 5.6.1.1 Intel® SSE2 Data Movement Instructions	
5-20 5.6.1.2 Intel® SSE2 Packed Arithmetic Instructions	
5-20 5.6.1.3 Intel® SSE2 Logical Instructions	
5-21 5.6.1.4 Intel® SSE2 Compare Instructions	
5-21 5.6.1.5 Intel® SSE2 Shuffle and Unpack Instructions.	
5-21 5.6.1.6 Intel® SSE2 Conversion Instructions.	
5-21 5.6.2 Intel® SSE2 Packed Single Precision Floating-Point Instructions	
5-22 5.6.3 Intel® SSE2 128-Bit SIMD Integer Instructions	
5-22 5.6.4 Intel® SSE2 Cacheability Control and Ordering Instructions	
5-22 5.7 INTEL® SSE3 INSTRUCTIONS.	
5-23 5.7.1 Intel® SSE3 x87-FP Integer Conversion Instruction.	
5-23 5.7.2 Intel® SSE3 Specialized 128-bit Unaligned Data Load Instruction.	
5-23 5.7.3 Intel® SSE3 SIMD Floating-Point Packed ADD/SUB Instructions.	
5-23 5.7.4 Intel® SSE3 SIMD Floating-Point Horizontal ADD/SUB Instructions	
5-23 5.7.5 Intel® SSE3 SIMD Floating-Point LOAD/MOVE/DUPLICATE Instructions	
5-24 5.7.6 Intel® SSE3 Agent Synchronization Instructions.	
5-24 5.8 SUPPLEMENTAL STREAMING SIMD EXTENSIONS 3 (SSSE3) INSTRUCTIONS	

5-24 5.8.1 Horizontal Addition/Subtraction	5-24 5.8.2 Packed Absolute Values	5-25 5.8.3 Multiply and Add Packed Signed and Unsigned Bytes	5-25 5.8.4 Packed Multiply High with Round and Scale
5-25 5.8.5 Packed Shuffle Bytes	5-25 5.8.6 Packed Sign	5-25 5.8.7 Packed Align Right	5-26 5.9 INTEL® SSE4 INSTRUCTIONS
5-26 5.10 INTEL® SSE4.1 INSTRUCTIONS	5-26 5.10.1 Dword Multiply Instructions	5-26 5.10.2 Floating-Point Dot Product Instructions	5-27 5.10.3 Streaming Load Hint Instruction
5-27 5.10.4 Packed Blending Instructions	5-27 5.10.5 Packed Integer MIN/MAX Instructions	5-27 5.10.6 Floating-Point Round Instructions with Selectable Rounding Mode	5-27 5.10.7 Insertion and Extractions from XMM Registers
5-28 5.10.8 Packed Integer Format Conversions	5-28 5.10.9 Improved Sums of Absolute Differences (SAD) for 4-Byte Blocks	5-28	

vi Vol. 1

CONTENTS

PAGE

5.10.10 Horizontal Search	5-29 5.10.11 Packed Test	5-29 5.10.12 Packed Qword Equality Comparisons	5-29 5.10.13 Dword Packing With Unsigned Saturation
5-29 5.11 INTEL® SSE4.2 INSTRUCTION SET	5-29 5.11.1 String and Text Processing Instructions	5-29 5.11.2 Packed Comparison SIMD integer Instruction	5-29 5.12 INTEL® AES-NI AND PCLMULQDQ
5-29 5.13 INTEL® ADVANCED VECTOR EXTENSIONS (INTEL® AVX)	5-30 5.14 16-BIT FLOATING-POINT CONVERSION	5-30 5.15 FUSED-MULTIPLY-ADD (FMA)	5-30 5.16 INTEL® ADVANCED VECTOR EXTENSIONS 2 (INTEL® AVX2)
5-31 5.17 INTEL® TRANSACTIONAL SYNCHRONIZATION EXTENSIONS (INTEL® TSX)	5-31 5.18 INTEL® SHA EXTENSIONS	5-31 5.19 INTEL® ADVANCED VECTOR EXTENSIONS 512 (INTEL® AVX-512)	5-31 5.20 SYSTEM INSTRUCTIONS
5-36 5.21 64-BIT MODE INSTRUCTIONS	5-37 5.22 VIRTUAL-MACHINE EXTENSIONS	5-38 5.23 SAFER MODE EXTENSIONS	5-38 5.24 INTEL® MEMORY PROTECTION EXTENSIONS
5-39 5.25 INTEL® SOFTWARE GUARD EXTENSIONS	5-39 5.26 SHADOW STACK MANAGEMENT INSTRUCTIONS	5-40 5.27 CONTROL TRANSFER TERMINATING INSTRUCTIONS	5-40 5.28 INTEL® AMX INSTRUCTIONS
5-40 5.29 USER INTERRUPT INSTRUCTIONS	5-40 5.30 ENQUEUE STORE INSTRUCTIONS	5-41	

CHAPTER 6

PROCEDURE CALLS, INTERRUPTS, AND EXCEPTIONS

6.1 PROCEDURE CALL TYPES

6-1 6.2 STACKS	
6-1 6.2.1 Setting Up a Stack	
6-2 6.2.2 Stack Alignment	
6-2 6.2.3 Address-Size Attributes for Stack Accesses	
6-3 6.2.4 Procedure Linking Information	
6-3 6.2.4.1 Stack-Frame Base Pointer	
6-3 6.2.4.2 Return Instruction Pointer	
6-3 6.2.5 Stack Behavior in 64-Bit Mode	
6-4 6.3 SHADOW STACKS	
6-4 6.4 CALLING PROCEDURES USING CALL AND RET	
6-4 6.4.1 Near CALL and RET Operation	
6-4 6.4.2 Far CALL and RET Operation	
6-5 6.4.3 Parameter Passing	
6-7 6.4.3.1 Passing Parameters Through the General-Purpose Registers	
6-7 6.4.3.2 Passing Parameters on the Stack	
6-7 6.4.3.3 Passing Parameters in an Argument List	
6-7 6.4.4 Saving Procedure State Information	
6-7 6.4.5 Calls to Other Privilege Levels	
6-4.6 CALL and RET Operation Between Privilege Levels	
6-8 6.4.7 Branch Functions in 64-Bit Mode	
6-12 6.5 INTERRUPTS AND EXCEPTIONS	
6-12 6.5.1 Call and Return Operation for Interrupt or Exception Handling Procedures	
6-13 6.5.2 Calls to Interrupt or Exception Handler Tasks	
6-18 6.5.3 Interrupt and Exception Handling in Real-Address Mode	
6-18 6.5.4 INT n, INTO, INT3, INT1, and BOUND Instructions	
6-18 6.5.5 Handling Floating-Point Exceptions	
6-19 6.5.6 Interrupt and Exception Behavior in 64-Bit Mode	
6-19 6.6 PROCEDURE CALLS FOR BLOCK-STRUCTURED LANGUAGES	
6-20 6.6.1 ENTER Instruction	
6-20	

CONTENTS

PAGE

6.6.2 LEAVE Instruction	
6-24	

CHAPTER 7

PROGRAMMING WITH GENERAL-PURPOSE INSTRUCTIONS

7.1 PROGRAMMING ENVIRONMENT FOR GP INSTRUCTIONS	
7-1 7.2 PROGRAMMING ENVIRONMENT FOR GP INSTRUCTIONS IN 64-BIT MODE	
7-1 7.3 SUMMARY OF GP INSTRUCTIONS	
7-2 7.3.1 Data Transfer Instructions	
7-2 7.3.1.1 General Data Movement Instructions	
7-3 7.3.1.2 Exchange Instructions	
7-4 7.3.1.3 Exchange Instructions in 64-Bit Mode	
7-5 7.3.1.4 Stack Manipulation Instructions	
7-5 7.3.1.5 Stack Manipulation Instructions in 64-Bit Mode	
7-7 7.3.1.6 Type Conversion Instructions	
7-7 7.3.1.7 Type Conversion Instructions in 64-Bit Mode	
7-8 7.3.2 Binary Arithmetic Instructions	
7-8 7.3.2.1 Addition and Subtraction Instructions	
7-8 7.3.2.2 Increment and Decrement Instructions	
7-8 7.3.2.3 Increment and Decrement Instructions in 64-Bit Mode	
7-8 7.3.2.4 Comparison and Sign Change Instructions	

.....7-8 7.3.2.5 Multiplication and Division Instructions.	
.....7-9 7.3.3 Decimal Arithmetic Instructions.	
.....7-9 7.3.3.1 Packed BCD Adjustment Instructions	
.....7-9 7.3.3.2 Unpacked BCD Adjustment Instructions	
.....7-9 7.3.4 Decimal Arithmetic Instructions in	
64-Bit Mode 7-10 7.3.5 Logical Instructions	
..... 7-10 7.3.6	
Shift and Rotate Instructions	
..... 7-10 7.3.6.1 Shift Instructions	
..... 7-10 7.3.6.2 Double-Shift Instructions.	
..... 7-12 7.3.6.3 Rotate Instructions	
..... 7-13 7.3.7 Bit and Byte Instructions	
..... 7-13 7.3.7.1 Bit Test and Modify Instructions	
..... 7-14 7.3.7.2 Bit Scan Instructions .	
..... 7-14 7.3.7.3 Byte	
Set on Condition Instructions.	
7-14 7.3.7.4 Test Instruction	
..... 7-14 7.3.8 Control Transfer Instructions	
..... 7-14 7.3.8.1 Unconditional Transfer Instructions.	
..... 7-14 7.3.8.2 Conditional Transfer Instructions	
..... 7-15 7.3.8.3 Control Transfer Instructions in 64-Bit Mode	
..... 7-17 7.3.8.4 Software Interrupt Instructions	
..... 7-17 7.3.8.5 Software Interrupt	
Instructions in 64-bit Mode and Compatibility Mode. 7-18 7.3.9 String	
Operations	
... 7-18 7.3.9.1 String Instructions	
..... 7-18 7.3.9.2 Repeated String Operations	
..... 7-19 7.3.9.3 Fast-String Operation	
..... 7-19 7.3.9.4 String Operations in 64-Bit Mode	
..... 7-20 7.3.10 I/O Instructions	
..... 7-20 7.3.11 I/O Instructions in	
64-Bit Mode 7-20 7.3.12	
Enter and Leave Instructions	
..... 7-21 7.3.13 Flag Control (EFLAGS) Instructions	
..... 7-21 7.3.13.1 Carry and Direction Flag Instructions.	
..... 7-21 7.3.13.2 EFLAGS Transfer Instructions	
..... 7-21 7.3.13.3 Interrupt Flag Instructions	
..... 7-22 7.3.14 Flag Control (RFLAGS) Instructions in	
64-Bit Mode 7-22 7.3.15 Segment Register	
Instructions	
7.3.15.1 Segment-Register Load and Store Instructions	
..... 7-22 7.3.15.2 Far Control Transfer Instructions	
..... 7-22 7.3.15.3 Software Interrupt Instructions	
..... 7-23 7.3.15.4 Load Far Pointer Instructions.	
..... 7-23	

7.3.16 Miscellaneous Instructions	
..... 7-23 7.3.16.1 Address Computation Instruction.	
..... 7-23 7.3.16.2 Table Lookup Instructions.	
..... 7-23 7.3.16.3 Processor Identification Instruction.	
..... 7-23 7.3.16.4 No-Operation and Undefined Instructions	
..... 7-23 7.3.17 Random Number Generator	
Instructions 7-24 7.3.17.1 RDRAND	
7-24 7.3.17.2 RDSEED.	
..... 7-24	

PROGRAMMING WITH THE X87 FPU

8.1 X87 FPU EXECUTION ENVIRONMENT	
8-1 8.1.1 x87 FPU in 64-Bit Mode and Compatibility Mode	
8-1 8.1.2 x87 FPU Data Registers	
8-1 8.1.2.1 Parameter Passing With the x87 FPU Register Stack	
8-3 8.1.3 x87 FPU Status Register	
8-4 8.1.3.1 Top of Stack (TOP) Pointer	
8-4 8.1.3.2 Condition Code Flags	
8-4 8.1.3.3 x87 FPU Floating-Point Exception Flags	
8-5 8.1.3.4 Stack Fault Flag	
8-6 8.1.4 Branching and Conditional Moves on Condition Codes	
8-6 8.1.5 x87 FPU Control Word	
8-7 8.1.5.1 x87 FPU Floating-Point Exception Mask Bits	
8-7 8.1.5.2 Precision Control Field	
8-7 8.1.5.3 Rounding Control Field	
8-8 8.1.6 Infinity Control Flag	
8-8 8.1.7 x87 FPU Tag Word	
8-8 8.1.8 x87 FPU Instruction and Data (Operand) Pointers	
8-9 8.1.9 Last Instruction Opcode	
8-10 8.1.9.1 Fopcode Compatibility Sub-mode	
8-10 8.1.10 Saving the x87 FPU State with FSTENV/FNSTENV and FSAVE/FNSAVE	
8-11 8.1.11 Saving the x87 FPU State with FXSAVE	
8-12 8.2 X87 FPU DATA TYPES	
8-13 8.2.1 Indefinites	
8-14 8.2.2 Unsupported Double Extended Precision Floating-Point Encodings and Pseudo-Denormals	
8-14 8.3 X87 FPU INSTRUCTION SET	
8-15 8.3.1 Escape (ESC) Instructions	
8-15 8.3.2 x87 FPU Instruction Operands	
8-15 8.3.3 Data Transfer Instructions	
8-15 8.3.4 Load Constant Instructions	
8-17 8.3.5 Basic Arithmetic Instructions	
8-17 8.3.6 Comparison and Classification Instructions	
8-18 8.3.6.1 Branching on the x87 FPU Condition Codes	
8-20 8.3.7 Trigonometric Instructions	
8-20 8.3.8 Approximation of Pi	
8-21 8.3.9 Logarithmic, Exponential, and Scale	
8-21 8.3.10 Transcendental Instruction Accuracy	
8-21 8.3.11 x87 FPU Control Instructions	
8-23 8.3.12 Waiting vs. Non-waiting Instructions	
8-24 8.3.13 Unsupported x87 FPU Instructions	
8-24 8.4 X87 FPU FLOATING-POINT EXCEPTION HANDLING	
8-24 8.4.1 Arithmetic vs. Non-arithmetic Instructions	
8-25 8.5 X87 FPU FLOATING-POINT EXCEPTION CONDITIONS	
8-26 8.5.1 Invalid Operation Exception	
8-26 8.5.1.1 Stack Overflow or Underflow Exception (#IS)	
8-26 8.5.1.2 Invalid Arithmetic Operand Exception (#IA)	
8-27 8.5.2 Denormal Operand Exception (#D)	
8-28 8.5.3 Divide-By-Zero Exception (#Z)	
8-28 8.5.4 Numeric Overflow Exception (#O)	
8-28	

8.5.5 Numeric Underflow Exception (#U)	8-29
8.5.6 Inexact-Result (Precision) Exception (#P)	8-30
8.6 X87 FPU EXCEPTION SYNCHRONIZATION	8-30
8.7 HANDLING X87 FPU EXCEPTIONS IN SOFTWARE	8-31
8.7.1 Native Mode.	8-32
8.7.2 MS-DOS* Compatibility Sub-mode.	8-32
8.7.3 Handling x87 FPU Exceptions in Software	8-33

CHAPTER 9

PROGRAMMING WITH INTEL® MMX™ TECHNOLOGY

9.1 OVERVIEW OF MMX TECHNOLOGY	9-1
9.2 THE MMX TECHNOLOGY PROGRAMMING ENVIRONMENT	9-2
9.2.1 MMX Technology in 64-Bit Mode and Compatibility Mode	9-2
9.2.2 MMX Registers	9-2
9.2.3 MMX Data Types	9-3
9.2.4 Memory Data Formats	9-3
9.2.5 Single Instruction, Multiple Data (SIMD) Execution Model	9-4
9.3 SATURATION AND WRAPAROUND MODES	9-4
9.4 MMX INSTRUCTIONS	9-5
9.4.1 Data Transfer Instructions	9-6
9.4.2 Arithmetic Instructions.	9-6
9.4.3 Comparison Instructions.	9-7
9.4.4 Conversion Instructions	9-7
9.4.5 Unpack Instructions.	9-7
9.4.6 Logical Instructions	9-7
9.4.7 Shift Instructions	9-8
9.4.8 EMMS Instruction	9-8
9.5 COMPATIBILITY WITH X87 FPU ARCHITECTURE	9-8
9.5.1 MMX Instructions and the x87 FPU Tag Word	9-8
9.6 WRITING APPLICATIONS WITH MMX CODE.	9-8
9.6.1 Checking for MMX Technology Support	9-8
9.6.2 Transitions Between x87 FPU and MMX Code	9-9
9.6.3 Using the EMMS Instruction	9-9
9.6.4 Mixing MMX and x87 FPU Instructions	9-10
9.6.5 Interfacing with MMX Code.	9-10
9.6.6 Using MMX Code in a Multitasking Operating System Environment.	9-10
9.6.7 Exception Handling in MMX Code	9-11
9.6.8 Register Mapping	9-11
9.6.9 Effect of Instruction Prefixes on MMX Instructions	9-11

CHAPTER 10

PROGRAMMING WITH INTEL® STREAMING SIMD EXTENSIONS (INTEL® SSE)

10.1 OVERVIEW OF INTEL® SSE	10-1
10.2 INTEL® SSE PROGRAMMING ENVIRONMENT	10-2
10.2.1 Intel® SSE in 64-Bit Mode and Compatibility Mode.	10-3
10.2.2 XMM Registers	10-3
10.2.3 MXCSR Control and Status Register.	10-3
10.2.3.1 SIMD Floating-Point Mask and Flag Bits	10-4
10.2.3.2 SIMD Floating-Point Rounding Control Field	10-4
10.2.3.3 Flush-To-Zero	10-4
10.2.3.4 Denormals-Are-Zeros	10-5
10.2.4 Compatibility of Intel® SSE with Intel® SSE2 and SSE3, MMX, and the x87 FPU	10-5
10.3 INTEL® SSE DATA TYPES	10-5
10.4 INTEL® SSE	10-5

INSTRUCTION SET	
10-6 10.4.1 Intel® SSE Packed and Scalar Floating-Point Instructions	
10-6 10.4.1.1 Intel® SSE Data Movement Instructions	
10-7 10.4.1.2 Intel® SSE Arithmetic Instructions	
10-8 10.4.2 Intel® SSE Logical Instructions	
10-9 10.4.2.1 Intel® SSE Comparison Instructions	
10-9	

x Vol. 1

CONTENTS

PAGE

10.4.2.2 Intel® SSE Shuffle and Unpack Instructions	
10-9 10.4.3 Intel® SSE Conversion Instructions	
10-11 10.4.4 Intel® SSE 64-Bit SIMD Integer Instructions	
10-11 10.4.5 MXCSR State Management Instructions	
10-12 10.4.6 Cacheability Control, Prefetch, and Memory Ordering Instructions	
10-12 10.4.6.1 Cacheability Control Instructions	
10-12 10.4.6.2 Caching of Temporal vs. Non-Temporal Data	
10-12 10.4.6.3 PREFETCHH Instructions	
10-13 10.4.6.4 SFENCE Instruction	
10-14 10.5 FXSAVE AND FXRSTOR INSTRUCTIONS	
10-14 10.5.1 FXSAVE Area	
10-14 10.5.1.1 x87 State	
10-15 10.5.1.2 SSE State	
10-16 10.5.2 Operation of FXSAVE	
10-16 10.5.3 Operation of FXRSTOR	
10-17 10.6 HANDLING INTEL® SSE INSTRUCTION EXCEPTIONS	
10-17 10.7 WRITING APPLICATIONS WITH INTEL® SSE	
10-17	

CHAPTER 11

PROGRAMMING WITH INTEL® STREAMING SIMD EXTENSIONS 2 (INTEL® SSE2)

11.1 OVERVIEW OF INTEL® SSE2	
11-1 11.2 INTEL® SSE2 PROGRAMMING ENVIRONMENT	
11-2 11.2.1 Intel® SSE2 in 64-Bit Mode and Compatibility Mode	
11-3 11.2.2 Compatibility of Intel® SSE2 with Intel® SSE, MMX Technology, and x87 FPU Programming Environment	
11-3 11.2.3 Denormals-Are-Zeros Flag	
11-3 11.3 INTEL® SSE2 DATA TYPES	
11.4 INTEL® SSE2 INSTRUCTIONS	
11-4 11.4.1 Packed and Scalar Double Precision Floating-Point Instructions	
11-4 11.4.1.1 Data Movement Instructions	
11-5 11.4.1.2 Intel® SSE2 Arithmetic Instructions	
11-6 11.4.1.3 Intel® SSE2 Logical Instructions	
11-7 11.4.1.4 Intel® SSE2 Comparison Instructions	
11-7 11.4.1.5 Intel® SSE2 Shuffle and Unpack Instructions	
11-9 11.4.1.6 Intel® SSE2 Conversion Instructions	
11-11 11.4.2 Intel® SSE2 64-Bit and 128-Bit SIMD Integer Instructions	
11-11 11.4.3 128-Bit SIMD Integer Instruction Extensions	
11-12 11.4.4 Cacheability Control and Memory Ordering Instructions	
11-12 11.4.4.1 FLUSH Cache Line	
11-12 11.4.4.2 Cacheability Control Instructions	
11-12 11.4.4.3 Memory Ordering Instructions	
11-12 11.4.4.4 Pause	
11-12 11.4.5 Branch Hints	
11-13 11.5 INTEL® SSE, SSE2, AND SSE3 EXCEPTIONS	
11-13 11.5.1 SIMD Floating-Point Exceptions	

Conditions	11-13	11.5.2 SIMD Floating-Point Exception
Operation Exception (#I)	11-14	11.5.2.1 Invalid
11.5.2.2 Denormal-Operand Exception (#D)	11-15	
11.5.2.3 Divide-By-Zero Exception (#Z)	11-15	
11.5.2.4 Numeric Overflow Exception (#O)	11-15	
11.5.2.5 Numeric Underflow Exception (#U)	11-16	
11.5.2.6 Inexact-Result (Precision) Exception (#P)	11-16	
Exceptions	11-16	11.5.3 Generating SIMD Floating-Point
Masked Exceptions	11-17	11.5.3.1 Handling
11.5.3.2 Handling Unmasked Exceptions	11-18	
11.5.3.3 Handling Combinations of Masked and Unmasked Exceptions	11-18	
11.5.4 Handling SIMD Floating-Point Exceptions in Software.	11-18	
11.5.5 Interaction of SIMD and x87 FPU Floating-Point Exceptions.	11-18	
11.6 WRITING APPLICATIONS WITH INTEL® SSE AND SSE2	11-19	
11.6.1 General Guidelines for Using Intel® SSE and SSE2	11-19	

Vol. 1 xi

CONTENTS

PAGE

11.6.2 Checking for Intel® SSE and SSE2 Support	11-19
11.6.3 Checking for the DAZ Flag in the MXCSR Register	11-20
11.6.4 Initialization of Intel® SSE and SSE2	11-20
11.6.5 Saving and Restoring the SSE/SSE2 State.	11-21
11.6.6 Guidelines for Writing to the MXCSR Register	11-21
11.6.7 Interaction of Intel® SSE and SSE2 Instructions with x87 FPU and MMX Instructions.	11-22
11.6.8 Compatibility of SIMD and x87 FPU Floating-Point Data Types	11-22
11.6.9 Mixing Packed and Scalar Floating-Point and 128-Bit SIMD Integer Instructions and Data	11-22
11.6.10 Interfacing with Intel® SSE and SSE2 Procedures and Functions	11-23
11.6.10.1 Passing Parameters in XMM Registers	11-23
11.6.10.2 Saving XMM Register State on a Procedure or Function Call.	11-23
11.6.10.3 Caller-Save Recommendation for Procedure and Function Calls.	11-24
11.6.11 Updating Existing MMX Technology Routines Using 128-Bit SIMD Integer Instructions.	11-24
11.6.12 Branching on Arithmetic Operations	11-24
11.6.13 Cacheability Hint Instructions.	11-25
11.6.14 Effect of Instruction Prefixes on Intel® SSE and SSE2 Instructions	11-25

CHAPTER 12

PROGRAMMING WITH INTEL® SSE3, SSSE3, INTEL® SSE4, AND INTEL® AES-NI

12.1 PROGRAMMING ENVIRONMENT AND DATA TYPES.	12-1
12.1.1 Intel® SSE3, SSSE3, and Intel® SSE4 in 64-Bit Mode and Compatibility Mode	12-1
12.1.2 Compatibility of Intel® SSE3 and SSSE3 with MMX Technology, the x87 FPU Environment, and Intel® SSE and SSE2	12-1
12.1.3 Horizontal and Asymmetric Processing.	12-1
12.2 OVERVIEW OF INTEL® SSE3 INSTRUCTIONS	12-2
12.3 INTEL® SSE3 INSTRUCTIONS.	12-2
12.3.1 x87 FPU Instruction for Integer Conversion	12-3
12.3.2 SIMD Integer Instruction for Specialized 128-bit Unaligned Data Load	12-3
12.3.3 SIMD Floating-Point Instructions That Enhance LOAD/MOVE/DUPLICATE Performance.	12-3
12.3.4 SIMD Floating-Point Instructions Provide Packed Addition/Subtraction.	12-4
12.3.5 SIMD Floating-Point Instructions Provide Horizontal Addition/Subtraction.	12-4
12.3.6 Two Thread Synchronization Instructions	12-5
12.4 WRITING APPLICATIONS WITH INTEL® SSE3	12-5
12.4.1 Guidelines for Using Intel® SSE3	12-5
12.4.2 Checking for Intel® SSE3 Support	12-5

12.4.3 Enable FTZ and DAZ for SIMD Floating-Point Computation	
12-6 12.4.4 Programming Intel® SSE3 with Intel® SSE and SSE2	
12-6 12.5 OVERVIEW OF SSSE3 INSTRUCTIONS	
12-6 12.6 SSSE3 INSTRUCTIONS	
12-6 12.6.1 Horizontal Addition/Subtraction	
12-7 12.6.2 Packed Absolute Values	12-7
12.6.3 Multiply and Add Packed Signed and Unsigned Bytes	
12-8 12.6.4 Packed Multiply High with Round and Scale	
12-8 12.6.5 Packed Shuffle Bytes	
12-8 12.6.6 Packed Sign	
12-8 12.6.7 Packed Align Right	
12-8 12.7 WRITING APPLICATIONS WITH SSSE3 EXTENSIONS	
12-9 12.7.1 Guidelines for Using SSSE3	
12-9 12.7.2 Checking for SSSE3 Support	
12-9 12.8 INTEL® SSE3, SSSE3, AND INTEL® SSE4 EXCEPTIONS	
12-9 12.8.1 Device Not Available (DNA) Exceptions	
12-9 12.8.2 Numeric Error flag and IGNNE#	12-9 12.8.3
Emulation	
12-10 12.8.4 IEEE 754 Compliance of Intel® SSE4.1 Floating-Point Instructions	
12-10 12.9 INTEL® SSE4 OVERVIEW	
12-10 12.10 INTEL® SSE4.1 INSTRUCTION SET	
12-11 12.10.1 Dword Multiply Instructions	
12-11 12.10.2 Floating-Point Dot Product Instructions	12-11 12.10.3
Streaming Load Hint Instruction	
12-12 12.10.4 Packed Blending Instructions	
12-14	

12.10.5 Packed Integer MIN/MAX Instructions	
12-14 12.10.6 Floating-Point Round Instructions with Selectable Rounding Mode	
12-14 12.10.7 Insertion and Extractions from XMM Registers	
12-15 12.10.8 Packed Integer Format Conversions	
12-15 12.10.9 Improved Sums of Absolute Differences (SAD) for 4-Byte Blocks	12-16
12-16 12.10.10 Horizontal Search	
12-16 12.10.11 Packed Test	
12-17 12.10.12 Packed Qword Equality Comparisons	
12-17 12.10.13 Dword Packing With Unsigned Saturation	
12-17 12.11 INTEL® SSE4.2 INSTRUCTION SET	
12-17 12.11.1 String and Text Processing Instructions	
12-17 12.11.1.1 Memory Operand Alignment	
12-18 12.11.2 Packed Comparison SIMD Integer Instruction	
12-18 12.12 WRITING APPLICATIONS WITH INTEL® SSE4 EXTENSIONS	
12-18 12.12.1 Guidelines for Using Intel® SSE4 Extensions	
12-18 12.12.2 Checking for Intel® SSE4.1 Support	
12-19 12.12.3 Checking for Intel® SSE4.2 Support	
12-19 12.13 INTEL® AES-NI OVERVIEW	
12-19 12.13.1 Little-Endian Architecture and Big-Endian Specification (FIPS 197)	
12-19 12.13.1.1 AES Data Structure in Intel® 64 Architecture	
12-20 12.13.2 AES Transformations and Functions	
12-21 12.13.3 PCLMULQDQ	
12-24 12.13.4 Checking for Intel® AES-NI Support	
12-24	

CHAPTER 13

MANAGING STATE USING THE XSAVE FEATURE SET

13.1 XSAVE-SUPPORTED FEATURES AND STATE-COMPONENT BITMAPS.	13-1
13.2 ENUMERATION OF CPU SUPPORT FOR XSAVE INSTRUCTIONS AND XSAVE-SUPPORTED FEATURES	13-3
13.3 ENABLING THE XSAVE FEATURE SET AND XSAVE-ENABLED FEATURES.	13-5
13.4 XSAVE AREA	13-7
13.4.1 Legacy Region of an XSAVE Area	13-8
13.4.2 XSAVE Header	13-9
13.4.3 Extended Region of an XSAVE Area	13-9
13.5 XSAVE-MANAGED STATE	13-10
13.5.1 x87 State	13-10
13.5.2 SSE State	13-11
13.5.3 AVX State	13-11
13.5.4 MPX State	13-11
13.5.5 AVX-512 State.	13-12
13.5.6 PT State	13-13
13.5.7 PKRU State	13-13
13.5.8 PASID State	13-14
13.5.9 CET State	13-14
13.5.10 HDC State.	13-14
13.5.11 UINTR State.	13-15
13.5.12 LBR State	13-16
13.5.13 HWP State	13-17
13.5.14 AMX State	13-17
13.6 PROCESSOR TRACKING OF XSAVE-MANAGED STATE	13-18
13.7 OPERATION OF XSAVE.	13-19
13.8 OPERATION OF XRSTOR	13-20
13.8.1 Standard Form of XRSTOR	13-20
13.8.2 Compacted Form of XRSTOR	13-21
13.8.3 XRSTOR and the Init and Modified Optimizations	13-21
13.9 OPERATION OF XSAVEOPT	13-22
13.10 OPERATION OF XSAVEC	13-23
13.11 OPERATION OF XSAVES	13-24
13.12 OPERATION OF XRSTORS	13-25
13.13 MEMORY ACCESSES BY THE XSAVE FEATURE SET	13-27

Vol. 1 xiii

CONTENTS

PAGE

13.14 EXTENDED FEATURE DISABLE (XFD).	13-27
---	-------

CHAPTER 14

PROGRAMMING WITH INTEL® AVX, FMA, AND INTEL® AVX2

14.1 INTEL® AVX OVERVIEW	14-1
14.1.1 256-Bit Wide SIMD Register Support.	14-1
14.1.2 Instruction Syntax Enhancements	14-2
14.1.3 VEX Prefix Instruction Encoding Support.	14-2
14.2 FUNCTIONAL OVERVIEW	14-3
14.2.1 256-bit Floating-Point Arithmetic Processing Enhancements	14-9
14.2.2 256-bit Non-Arithmetic Instruction Enhancements	14-9
14.2.3 Arithmetic Primitives for 128-bit Vector and Scalar processing.	14-11
14.2.4 Non-Arithmetic Primitives for 128-bit Vector and Scalar Processing	14-11

14-13	14.3 DETECTION OF INTEL® AVX INSTRUCTIONS	14-13
14-15	14.3.1 Detection of VEX-Encoded AES and VPCLMULQDQ	14-15
14-17	14.4 HALF PRECISION FLOATING-POINT CONVERSION	14-17
14-18	14.4.1 Detection of F16C Instructions	14-18
14-20	14.5 FUSED-MULTIPLY-ADD (FMA) EXTENSIONS	14-20
14-21	14.5.1 FMA Instruction Operand Order and Arithmetic Behavior	14-21
14-22	14.5.2 Fused-Multiply-ADD (FMA) Numeric Behavior	14-22
14-22	14.5.3 Detection of FMA	14-22
14-24	14.6 OVERVIEW OF INTEL® ADVANCED VECTOR EXTENSIONS 2 (INTEL® AVX2)	14-24
14-25	14.6.1 AVX2 and 256-bit Vector Integer Processing	14-25
14-25	14.7 PROMOTED VECTOR INTEGER INSTRUCTIONS IN INTEL® AVX2	14-25
14-26	14.7.1 Detection of Intel® AVX2	14-26
14-31	14.8 ACCESSING YMM REGISTERS	14-31
14-32	14.9 MEMORY ALIGNMENT	14-32
14-32	14.10 SIMD FLOATING-POINT EXCEPTIONS	14-32
14-34	14.11 EMULATION	14-34
14-34	14.12 WRITING INTEL® AVX FLOATING-POINT EXCEPTION HANDLERS	14-34
14-34	14.13 GENERAL PURPOSE INSTRUCTION SET ENHANCEMENTS	14-34
14-35		14-35

CHAPTER 15

PROGRAMMING WITH INTEL® AVX-512

15-1	15.1 OVERVIEW	15-1
15-1	15.1.1 512-Bit Wide SIMD Register Support	15-1
15-1	15.1.2 32 SIMD Register Support	15-1
15-1	15.1.3 Eight Opmask Register Support	15-1
15-1	15.1.4 Instruction Syntax Enhancement	15-1
15-2	15.1.5 EVEX Instruction Encoding Support	15-2
15-3	15.2 DETECTION OF AVX-512 FOUNDATION INSTRUCTIONS	15-3
15-3	15.2.1 Additional 512-bit Instruction Extensions of the Intel AVX-512 Family	15-3
15-4	15.2.2 Detection of AVX512-FP16 Instructions	15-4
15-5	15.3 DETECTION OF 512-BIT INSTRUCTION GROUPS OF INTEL® AVX-512 FAMILY	15-5
15-6	15.4 DETECTION OF INTEL® AVX-512 INSTRUCTION GROUPS OPERATING AT 256 AND 128-BIT VECTOR LENGTHS	15-6
15-7	15.5 ACCESSING XMM, YMM, AND ZMM REGISTERS	15-7
15-8	15.6 ENHANCED VECTOR PROGRAMMING ENVIRONMENT USING EVEX ENCODING	15-8
15-8	15.6.1 OPMASK Register to Predicate Vector Data Processing	15-8
15-9	15.6.1.1 Opmask Register K0	15-9
15-9	15.6.1.2 Example of Opmask Usages	15-9
15-10	15.6.2 OpMask Instructions	15-10
15-11	15.6.3 Broadcast	15-11
15-11	15.6.4 Static Rounding Mode and Suppress All Exceptions	15-11
15-12	15.6.5 Compressed Disp8*N Encoding	15-12
15-13	15.7 MEMORY ALIGNMENT	15-13
15-13	15.8 SIMD FLOATING-POINT EXCEPTIONS	15-13
15-14	15.9 INSTRUCTION EXCEPTION SPECIFICATION	15-14
15-15	15.10 EMULATION	15-15
15-15	15.11 WRITING FLOATING-POINT EXCEPTION HANDLERS	15-15

CHAPTER 16

PROGRAMMING WITH INTEL® TRANSACTIONAL SYNCHRONIZATION EXTENSIONS